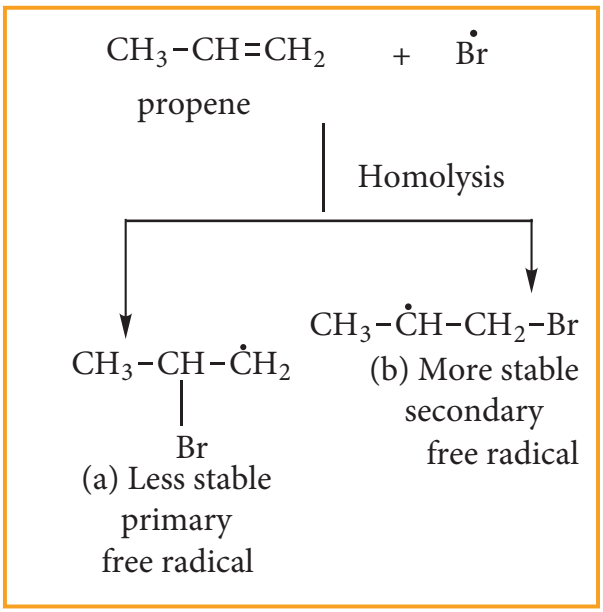
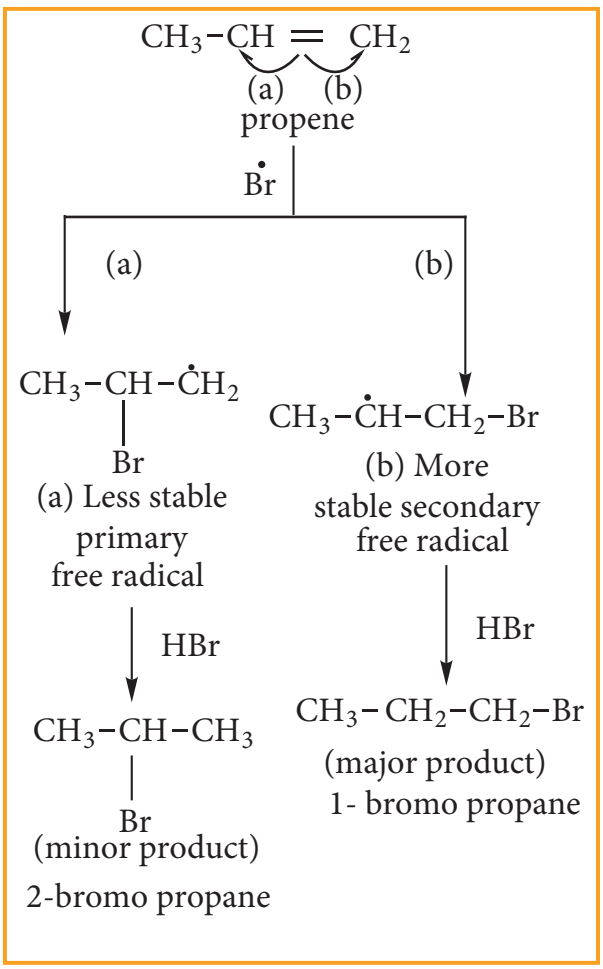
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Step:4

Addition of HBr to secondary free radical



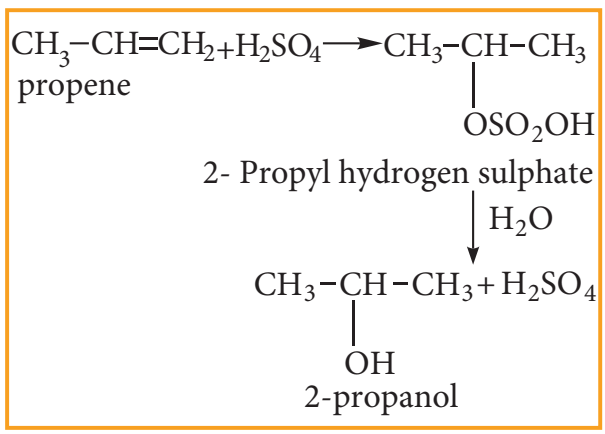
The H-Cl bond is stronger (430.5 kJmol^{-1}) than H-Br bond (363.7 kJmol^{-1}), thus H-Cl is not cleaved by the free radical. The H-I bond is weaker (296.8 kJ mol^{-1}), than H-Cl bond. Thus H-I bond breaks easily but iodine free radicals combine to form iodine molecules instead of adding to the double bond and hence peroxide effect is not observed in HCl& HI.

Kharasch Addition

Metal catalysed free radical addition of $CXCl_3$ Compounds to alkene is called Kharash addition reaction

(v) Addition of sulphuric acid to alkenes

Alkenes react with cold and concentrated sulphuric acid to form alkyl hydrogen sulphate accordance with Markownikoff 's rule. Further hydrolysis yields alcohol.



(2) Oxidation:

(i) With cold dilute alkaline KMnO4 solution (Baeyer's Reagent)

Alkenes react with Baeyer's reagent to form vicinal diols. The purple solution (Mn^{2+}) becomes dark green (Mn^{6+}), and then produces a dark brown precipitate (Mn^{4+}).